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Energy efficiency maximization in UAV communication networks with nonlinear energy harvesting

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ABSTRACT

Wireless power transfer (WPT) technology has great potential for future Internet of Things (IoT) application scenarios of energy-constrained unmanned aerial vehicle (UAV)-assisted communication. To fully obtain power multiplexing gain and improve spectral efficiency, this paper investigates nonorthogonal multiple access (NOMA)-based WPT-charging UAV communication networks, where a multi-antenna base station (BS) transmits radio frequency energy signals to UAVs, and UAVs harvest energy and send the data information back to the BS. Nevertheless, due to practical limitations and the difficulty of linear energy harvesting (EH) models to capture the end-to-end energy conversion of real EH circuits, the effects of nonlinear EH should be considered. Considering the great difference between nonlinear and linear EH models on the system energy efficiency (EE), the EE maximization problem is established by jointly optimizing multivariable. As the problem is nontrivial and nonconvex, we transform the original problem from sum-of-ratios form into an equivalent objective function in subtractive form, then solve it by Lagrange dual decomposition approach. The simulation results verify the feasibility of the proposed algorithm, and the EE gains are obtained compared with benchmarks when the nonlinear EH strategy is fully considered.

1. Introduction

Recently, unmanned aerial vehicles (UAVs) have enabled manifold applications in various scenarios, such as environmental and natural disaster monitoring, border surveillance, emergency assistance, search and rescue missions, and so on [1]. Especially for the usage of small-scale UAVs, it is a practical and fantastic choice for commercial applications due to their ease of deployment, low acquisition and maintenance costs, high-maneuverability and ability to hover [2]. The explosive growth of mobile IoT devices and ever-increasing demand for high data rates impose critical challenges for the beyond fifth generation (B5G) networks that involve ultrahigh data rates of UAV communications. Moreover, the power-constrained UAV devices and bandwidth-limited wireless channels have promoted the inevitable exploration of the high spectrum- and energy-efficient wireless communication systems [3].

To meet the requirements of high data rate and spectrum efficiency in UAV communication, some researchers have applied the nonorthogonal multiple access (NOMA) technique in UAV communication scenarios. Compared with the conventional orthogonal multiple access, NOMA enables multiple users to share the same physical resources (time/frequency/code) by employing superposition coding at the transmitters and successive interference cancellation (SIC) at the receivers, and thus achieves better spectrum utilization [4]. In [5], a UAV-assisted NOMA-mobile edge computing communication network architecture is studied to maximize the system EE by jointly optimizing the user scheduling, resource allocation and UAV flight trajectory. In [6], Mei et al. applied the NOMA to the uplink communication from a UAV to cellular base stations (BSs), under spectrum sharing with the existing ground users. Hanif et al. maximized the downlink sum rate of a multiple-input single-output NOMA system by optimizing the weighted precoding vectors [7]. In [8], Mouni et al. evaluated

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the performance of the UAV-NOMA system with imperfect SIC and short packet transmission. Masaracchia et al. proposed an optimization framework for power and time resource allocation during time sharing NOMA transmissions performed by a UAV, and maximized the energy efficiency (EE) and the downlink throughput fairness among users within the UAV communication range [9]. In [10], the problem of EE maximization of a UAV-assisted full-duplex NOMA system is investigated and a novel as well as less complex adaptive-geometric distribution scheme is proposed to adaptively maintain equal cluster size and allocate resource optimally among the users. All the works above involve performance gains such as spectral efficiency, high reliability and massive connectivity in NOMA-based UAV communication scenarios, however, the limitations of the onboard battery capacity and the flying range pose an important challenge to most applications while performing daily tasks such as mission rescue, air parcel delivery or aerial long-distance communications in large areas.

In addition to the data rate and spectrum challenges, one serious shortcoming of the UAV is lacking in energy, since a mission failure or even a sudden crash of the UAV may happen if the battery cannot guarantee the UAV to complete the mission [11]. Wireless power transfer (WPT) is regarded as an efficient technique that can decrease the energy cost of wireless networks, significantly reduce the network energy expenditure, and provide energy supply for battery-limited nodes (such as UAVs) through energy harvesting [12]. In [13], Zheng et al. proposed the WPT-aided UAV communication in cell-free massive multiple-input multiple-output (MIMO) networks, where the harvested energy from the downlink WPT is used to support both uplink data and pilot transmission. In [14], Chen et al. used the WPT technique (represented by the resonant beam charging) to supply the battery of short-endurance UAVs, and maximized the overall power transmission efficiency. In [15], Yan et al. took into account the power consumption of the UAV (power consumption during hovering and flight), the charging process from a base station to the UAV and the conversion loss of the energy harvester, and maximized the sum-energy received by all sensors to find the optimal strategy for UAV placement. In [16], Merabet et al. applied the WPT technique to wirelessly recharge UAVs on the fly and therefore to compute optimal trajectories for UAVs and extend their flying range. Khuong et al. analyzed both security and reliability performance in energy harvesting UAV communication systems via radio frequency (RF) signals [17]. The above works for the UAV energy harvesting (EH) process are based on a linear EH model, which however ignores the realistic impact of nonlinear EH process of practical circuits, causes severe resource allocation mismatches, and even leads to significant performance degradation for practical systems with highly nonlinear EH circuits [18]. As the conversion process of the received RF signal to a direct-current (DC) signal has been confirmed to be a nonlinear process, different received power levels will lead to significantly diverse conversion efficiencies.

Most existing studies have not yet considered the resource allocation problem of a charged UAV communication system based on a nonlinear EH model, nor have they considered the application of NOMA technology to improve the massive access of the Internet of Things (IoT) users and system energy efficiency in the environment of nonlinear EH and charging UAVs, thus motivating our investigation. Nevertheless, analyzing the problem of energy efficiency optimization under a nonlinear energy harvesting model brings huge challenges from the aspects of theoretical analysis, model complexity, optimization and simulation validation. With the traditional linear EH model, the corresponding constraints are usually linear and convex, which are relatively easy to handle [19]. However, nonlinear EH models can be highly complex and involve intricate mathematical formulations [20]. With the nonlinear EH model, the related constraints are nonconvex, which is difficult to deal with [18]. The optimization problem itself becomes more challenging when dealing with nonlinear models. Moreover, the validation of the model against real-world data or experimental results is essential for the reliability of any conclusions drawn from the analysis.

In this paper, we investigate the EE maximization problem in an NOMA-enabled WPT-charging UAV communication system with the nonlinear EH model. The main contributions are summarized as follows:

- We propose a new WPT-powered UAV NOMA communication system in which the UAVs use the downlink harvested energy to deliver the uplink data information back to the BS via the NOMA scheme. Especially, the nonlinear EH model is considered because of the factor that the energy harvesting of the actual circuit no longer increases linearly after reaching saturation.
- In view of the different impact between nonlinear and linear EH models on system EE, an EE maximization problem of joint transmit power and nonlinear EH time allocation is formulated. To solve this non-convex problem, we transform the original nonlinear fraction problem to a convex subtractive form, and then solve it by Lagrange dual decomposition.
- Simulation results verify the effectiveness of the proposed energy-efficient power and time allocation algorithm, also show that the EE performance of WPT-powered UAV networks under the proposed nonlinear EH model outperforms that under the linear EH model.

The rest of this work is organized as follows. The NOMA-based UAV system model with nonlinear energy harvesting is introduced in Section 2. In Section 3, the EE optimization problem is formulated and an iterative solution is proposed to solve the transformed equivalent problem. Section 4 shows the simulation results. Finally, the conclusions are drawn in Section 5.

2. System model

As shown in Fig. 1, we consider a NOMA-based UAV communication system, where a multiantenna BS serves I single-antenna rotary-ring UAVs. Herein, the UAVs act as aggregators (or aerial UAV users) to collect information for several scenarios (crowded sports venues, ground users whose direct links are blocked, blocked traffic flow information, and forest fire monitoring, and so on [21]) to provide timely disaster warnings and assist in speeding up rescue and recovery operations when the public communication network gets crippled [22]. The BS is equipped with a uniform rectangular array (URA) with M antennas and acts as a power transmitter as well as an information receiver. The UAV is the aerial user, denoted by $U_i (i \in 1, 2, \ldots, I)$. Along the URA plane of the BS, we set an orthonormal coordinate system with unit basis vectors $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$. The total number of antennas is $M = M_x M_y$, where M_x and M_y are the number of antennas in the x-axis and y-axis directions, respectively. Assuming the space between each antenna is d, the position of the BS antenna array is located at (x_0, y_0, z_0). The position of the ith UAV is located at (x_i, y_i, z_i). We assume that the position of the URA is fixed with the height dimension ignored and UAVs hover to collect data and transmit it back to the BS.

To achieve the rational allocation of time frame resources, a harvest-then-transmit protocol is exploited to prolong the endurance of the UAVs in MIMO NOMA networks. A communication cycle consists of two phases and total time of which is denoted by T. The first phase τ_i is the downlink wireless energy transmission, where the BS charges the UAVs to prolong the lifecycle of the UAVs. The second phase $(T - \tau_i)$ is the uplink wireless information transmission, where UAVs send the data information collected in the environment (e.g., forests, stadium, traffic and so on) up to the BS.

In the downlink, the channel between the URA and UAV is a typical air-to-ground channel, which contains strong line-of-sight (LoS), reflected nonline-of-sight (NLoS), and small-scale fading. In general, the influence of small-scale fading is smaller than LoS and NLoS [23], and the probability of NLoS can be neglected since the UAV has a higher altitude than the building surroundings. Therefore, we use the strong

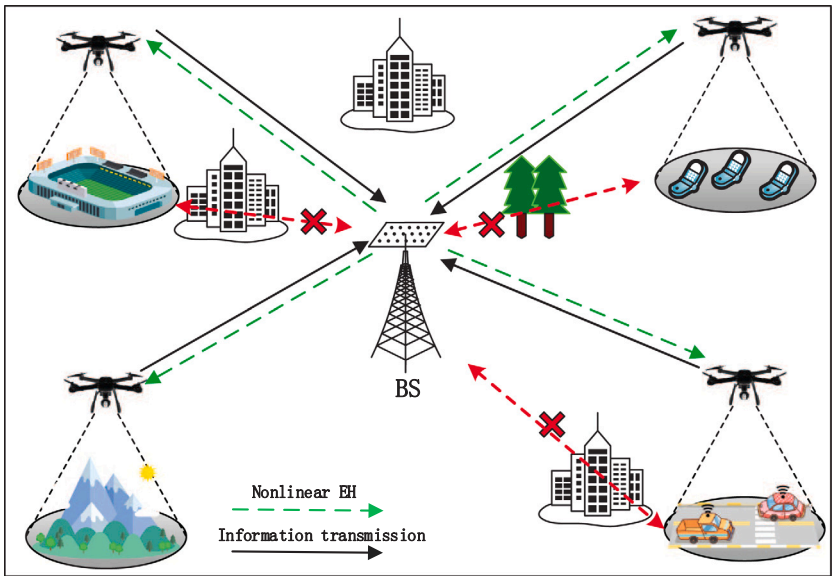


Fig. 1. A NOMA-based UAV communication network with nonlinear energy harvesting.

LoS model to denote the air-to-ground channel. As the UAVs are placed far from the BS and move in a small space, the relative differences in path loss are negligible. We normalize the downlink MIMO channel for U_i with respect to the distance. The channel between URA of the BS and U_i is denoted as

$$\mathbf{h}_i = \sqrt{M} v_i \varrho_{\mathrm{r}} \varrho_{\mathrm{t}} \mathbf{a}_{\mathrm{t}}\left(\phi_i, \theta_i\right), \tag{1}$$

where v_i is the channel gain depended on path loss, and $v_i \sim \mathcal{CN}\left(0, 10^{-0.1\mathrm{PL}(d_i)}\right)$; $\mathrm{PL}(d_i) = \varrho_a + 10\varrho_b \lg(d_i) + \xi$, and $\xi \sim \mathcal{N}\left(0, \sigma_\xi^2\right)^3$; d_i is the distance between the BS antenna array and U_i, and $d_i = \sqrt{\left(x_i - x_0\right)^2 + \left(y_i - y_0\right)^2 + \left(z_i - z_0\right)^2}$; ϱ_{r} and ϱ_{t} are the receive and transmit antenna element gains, respectively. $\mathbf{a}_{\mathrm{t}}\left(\phi_i, \theta_i\right)$ is the array steering vector of the BS, and $\mathbf{a}_{\mathrm{t}}(\phi_i, \theta_i) = \mathbf{a}_{\mathrm{tx}}^{\mathrm{az}}(\phi_i) \otimes \mathbf{a}_{\mathrm{ty}}^{\mathrm{el}}(\theta_i)$. $\mathbf{a}_{\mathrm{tx}}^{\mathrm{az}}(\phi_i) = \frac{1}{\sqrt{M_x}}\left[e^{-j\frac{2\pi d}{\lambda}\phi_i m_x}\right]_{m_x \in \mathcal{I}_x(M_x)}$, $\mathbf{a}_{\mathrm{ty}}^{\mathrm{el}}(\theta_i) = \frac{1}{\sqrt{M_y}}\left[e^{-j\frac{2\pi d}{\lambda}\theta_i m_y}\right]_{m_y \in \mathcal{I}_y(M_y)}$, $\mathcal{I}_x(M_x) = \{m_x - (M_x - 1)/2, m_x = 0, 1, \ldots, M_x - 1\}$, $\mathcal{I}_y(M_y) = \{m_y - (M_y - 1)/2, m_y = 0, 1, \ldots, M_y - 1\}$. λ is the wavelength of signal.

The downlink received RF power of U_i is written as

$$P_i = \tau_i \left|\mathbf{h}_i^{\mathrm{H}} \mathbf{a}_i\right|^2 p_i^t, \tag{2}$$

where p_i^t is the transmit power of the BS; $\mathbf{a}_i$ is the energy beamforming vector with $\mathbb{E}\left(\mathbf{a}_i^{\mathrm{H}} \mathbf{a}_i\right) = 1$; The classic and ideal linear EH model [24] can be expressed as

$$E_i^l = \epsilon P_i, \tag{3}$$

where ϵ is the energy conversion efficiency from the BS to electrical energy stored by U_i. We note that in this linear EH model, ϵ is independent of the transmit power level at each UAV. In other words, the total harvested energy at the UAVs is linearly and directly proportional to the received RF power. Nonetheless, the linear EH model is hard to capture the actual end-to-end energy conversion characteristics in the practical EH circuit. In this paper, we focus on power saturation and the correlation between input and output power without considering high sensitivity parameters [20,25]. For a better explanation of nonlinear EH impact and simplifying the mathematical analysis process, we model the nonlinear EH process as [26]

$$E_i^{nl} = \frac{\Lambda_i - c_i Y_i}{1 - Y_i}, \tag{4}$$

where c_i represents the maximum harvested power at the UAV when the EH reaches saturation. Λ_i is the traditional logistic function concerning the received RF power P_i, and Y_i is one of the variables of function maximum Λ_i, as given by

$$Y_i = \frac{1}{1 + \exp(a_i b_i)}, \tag{5}$$

$$\Lambda_i = \frac{Y_i}{1 + \exp\left(-a_i\left(P_i - b_i\right)\right)}, \tag{6}$$

respectively. Parameters a_i and b_i are the constants related to the detailed circuit specifications such as the resistance, capacitance, and diode turn-on voltage. The EH hardware circuit of each UAV is fixed, and a_i, b_i and c_i of the proposed model in (4)–(6) can be easily obtained according to standard curve fitting measurements [26].

In the uplink, each UAV sends the collected signals to the BS at either maximum power or collected transmit power. All the UAVs nonorthogonally transmit their signals with harvested power. Based on the reciprocity of uplink and downlink channels, the received signal at the BS is given by

$$\mathbf{y} = \sum_{i=1}^{I} \mathbf{h}_i x_i + \mathbf{n}, \tag{7}$$

where x_i is the information signal from the UAV to the BS, and $\mathbf{n}$ is the additive white Gaussian noise (AWGN) that obeys the standard normal distribution, i.e., $\mathbf{n} \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$. The uplink transmit power of U_i is given by

$$P_i^u = \frac{E_i^{nl}}{T - \tau_i}. \tag{8}$$

According to the principle of NOMA, one channel can be assigned to multiple users, who will use SIC to decode their signals [27]. Note that for the uplink NOMA, any information decoding order cannot affect the system sum rate or spectral efficiency [28]. We simply assume that the UAVs are decoded in descending order of their channel index, i.e., U_1 is first decoded, and $U_i (1 < i < I)$ is decoded afterwards, and U_I is finally decoded. Then, based on the Shannon capacity formula, the corresponding achievable rate of U_i is expressed as [29]

$$R_i = B \log_2\left(1 + M \frac{P_i^u |\mathbf{h}_i|^2}{\sum_{k=i+1}^{I} P_k^u |\mathbf{h}_k|^2 + \sigma^2}\right), \tag{9}$$

where B denotes the channel bandwidth and σ^2 denotes the noise power. The total channel throughput is given by

$$R_{\mathrm{tot}}\left(p_i^t, \tau_i\right) = \sum_{i=1}^{I} \left(T - \tau_i\right) R_i. \tag{10}$$

The total energy consumption of the NOMA-based UAV system is given by

$$E_{\text{tot}}\left(p_i^t, \tau_i\right) = \left(P_{\text{a}} N + I P_{\text{UAV}}\right) T + \sum_{i=1}^{I} p_i^t \tau_i, \tag{11}$$

where P_{a} is the power consumption of each antenna, and P_{UAV} is the power consumption of each rotary-wing UAV. $P_{\text{a}} = P_{\text{DAC}} + P_{\text{MIX}} + P_{\text{FILT}}$, where P_{DAC}, P_{MIX} and P_{FILT} are the power consumption of the digital-to-analog converter circuits, the mixer, and the transmit filter, respectively. P_{UAV} consists of two parts. One is the communication-related energy, i.e., the transmit power of signal processing P_{T} and the UAV antenna chain power consumption P_{a}. The other part is the propulsion energy, which is required for ensuring that the UAV remains aloft as well as for supporting its mobility. The propulsion power consumption model [30] for rotary-wing UAVs flying at a speed of V is denoted as $P(V) = P_0\left(1 + \frac{3V^2}{U_{\text{tip}}^2}\right) + P_i\left(\sqrt{1 + \frac{V^4}{4v_0^4}} - \frac{V^2}{2v_0^2}\right)^{\frac{1}{2}} + \frac{d_0 \rho s A V^3}{2}$, where P_0 and P_i are two constants related to the physical properties of UAV and the flight environment, such as weight, rotor radius and air density; U_{tip} denotes the tip speed of the rotor blade; v_0 is the mean rotor induced velocity during hovering; d_0 and s are the fuselage drag ratio and rotor solidity, respectively; ρ and A represent the air density and rotor disc area, respectively. The power required for hovering can be calculated as $P(0)$. Thus, $P_{\text{UAV}} = P_{\text{T}} + P_{\text{a}} + P(0)$.

The energy efficiency of the considered NOMA-based UAV systems with nonlinear EH is expressed as

$$\eta_{\text{EE}}(p_i^t, \tau_i) = \frac{R_{\text{tot}}\left(p_i^t, \tau_i\right)}{E_{\text{tot}}\left(p_i^t, \tau_i\right)}. \tag{12}$$

3. Problem formulation and solution

In this section, to improve the power multiplexing gain and provide maximum user access, we investigate the EE optimization problem under the scenario of NOMA-based WPT-powered UAV. Considering the nonlinear characteristic of the energy harvested by real circuits, we formulate the maximum EE problem based on the nonlinear EH model by jointly optimizing the EH time and transmit power of the BS. Due to the complexity of the nonlinear EH model and the nonconvexity of the formulated problem, we transform the original problem into a solvable problem, and propose an energy-efficient resource allocation algorithm to obtain the suboptimal solution.

3.1. Problem formulation

The EE maximization problem can be written as

$$\mathbf{P1}: \max_{\mathbf{p},\boldsymbol{\tau}} \ \eta_{\text{EE}}(p_i^t, \tau_i) \tag{13a}$$

$$\text{s.t. } \mathbf{C1}: \ 0 \le p_i^t \le p_{\text{bs,max}}, \tag{13b}$$

$$\mathbf{C2}: \ 0 \le P_i^u \le P_{\text{uav,max}}, \tag{13c}$$

$$\mathbf{C3}: \ 0 \le \tau_i \le T, \tag{13d}$$

$$\mathbf{C4}: \ R_i \ge R_{\min}, \tag{13e}$$

where $\mathbf{p} = [p_1^t, \ldots, p_i^t, \ldots, p_I^t]$ and $\boldsymbol{\tau} = [\tau_1, \ldots, \tau_i, \ldots, \tau_I]$ $(i \in [1, I])$ are the transmit power vector and WPT time vector, respectively. **C1** specifies the transmit power of the BS, which cannot be larger than the maximum transmit power $p_{\text{bs,max}}$ and be lower than 0; **C2** specifies the transmit power constraint of U_i, where $P_{\text{uav,max}}$ is the maximum power of U_i; **C3** specifies the time constraint of downlink nonlinear EH; **C4** specifies the minimum rate quality-of-service of the UAV, where $R_{\min}$ represents the minimum rate.

To effectively solve the formulated problem **P1**, we first substitute Eqs. (1)–(8) into **C2** and make the corresponding transformation, as given by

$$\tau_i \le \frac{-K_2 + \sqrt{K_2^2 - 4K_1K_3}}{2K_1} = \tau_{\max}, \tag{14}$$

where $K_1 = a_i \left|\mathbf{h}_i\right|^2 \mathbf{a}_i^2 P_{\text{bs,max}} P_{\text{uav,max}} e^{a_i b_i}$, $K_2 = P_{\text{uav,max}}\ a_i b_i + a_i c_i \left|\mathbf{h}_i\right|^2 \mathbf{a}_i^2 P_{\text{bs,max}} + a_i e^{a_i b_i} P_{\text{uav,max}} T \left|\mathbf{h}_i\right|^2 \mathbf{a}_i^2 P_{\text{bs,max}}$, $K_3 = [1 - (a_i b_i + 1)(c_i + e^{a_i b_i} P_{\text{uav,max}} T)]$.

3.2. Problem transformation and proposed algorithm

As the formulated problem in (13) is a non-convex fractional programming problem, we employ the Dinkelbach's method to convert **P1** into a subtractive form, as given by

$$\mathbf{P2}: \max_{\mathbf{p},\boldsymbol{\tau}} \ \Theta_{\text{EE}}\left(p_i^t, \tau_i\right) \tag{15a}$$

$$\text{s.t. } \mathbf{C1}, \mathbf{C}3, \mathbf{C4}, \tag{15b}$$

$$\widetilde{\mathbf{C}}2: \ \tau_i \le \tau_{\max}, \tag{15c}$$

where $\Theta_{\text{EE}}\left(p_i^t, \tau_i\right) = R_{\text{tot}}\left(p_i^t, \tau_i\right) - \eta_{\text{EE}}^* E_{\text{tot}}\left(p_i^t, \tau_i\right)$. The optimal EE value is denoted by η_{EE}^*, which can be written as

$$\eta_{\text{EE}}^* = \frac{\sum_{i=1}^{I}(T - \tau_i) R_i}{\left(P_{\text{a}} N + I P_{\text{uav}}\right) T + \sum_{i=1}^{I} p_i^t \tau_i} \mid p_i^t = p_i^{t*}, \tau_i = \tau_i^*, \tag{16}$$

where p_i^{t*} is the optimal transmit power and τ_i^* is the optimal WPT time slot. By Theorem 1, we can obtain an equivalent solution to **P1**.

Theorem 1. *The maximal EE η_{EE}^* can be achieved, if and only if*

$$\max_{\mathbf{p},\tau} R_{\text{tot}}\left(p_i^t, \tau_i\right) - \eta_{\text{EE}}^* E_{\text{tot}}\left(p_i^t, \tau_i\right) = 0 \tag{17}$$

Proof. See Appendix. ■

To effectively solve the problem **P2**, we can solve it by the Lagrange multiplier method and convex optimization tool to obtain its optimal solution in a dual domain. The Lagrange dual function of **P2** is written as

$$\begin{aligned} &\mathcal{L}\left(p_i^t, \tau_i, \kappa, \lambda, \mu, \nu\right) \\ &= R_{\text{tot}}\left(p_i^t, \tau_i\right) - \eta_{\text{EE}}^* E_{\text{tot}}\left(p_i^t, \tau_i\right) - \kappa\left(p_i^t - p_{\text{bs,max}}\right) \\ &- \lambda\left(\tau_i - \tau_{\max}\right) - \mu\left(\tau_i - T\right) - \nu\left(R_{\min} - R_i\right), \end{aligned} \tag{18}$$

where κ, λ, μ and ν are the positive Lagrange multipliers associated with the constraints of **P2**. Based on the analysis above, the dual problem of **P2** is expressed as

$$\mathbf{P3}: \min_{\kappa,\lambda,\mu,\nu} \max_{\mathbf{p},\boldsymbol{\tau}} \mathcal{L}\left(p_i^t, \tau_i, \kappa, \lambda, \mu, \nu\right). \tag{19}$$

We can obtain the optimal transmit power p_i^{t*} and optimal WPT time slot τ_i^* by solving the following Karush–Kuhn–Tucker (KKT) conditions:

$$\begin{aligned} &\frac{\partial \mathcal{L}\left(p_i^t, \tau_i, \kappa, \lambda, \mu, \nu\right)}{\partial p_i^t} = \left(T - \tau_i\right) \frac{\partial R_i}{\partial p_i^t} - \eta_{\text{EE}}^* \tau_i - \kappa + \nu \frac{\partial R_i}{\partial p_i^t} \\ &= \left(T - \tau_i + \nu\right) \frac{\partial R_i}{\partial p_i^t} - \eta_{\text{EE}}^* \tau_i - \kappa = 0, \end{aligned} \tag{20}$$

$$\frac{\partial \mathcal{L}\left(p_i^t, \tau_i, \kappa, \lambda, \mu, \nu\right)}{\partial \tau_i} = 0. \tag{21}$$

Further, we derive the optimal transmit power p_i^{t*} as follows

$$p_i^{t*} = \frac{1}{x_2} \log\left(\frac{-w_1 + \sqrt{w_1^2 - 4w_2 w_0}}{2w_2}\right), \tag{22}$$

where $w_0 = y_4 y_3 y_2^2$, $w_1 = y_4\left(2y_3 y_2^3 + Y y_2^2\right) - y_1$, $w_2 = y_4 y_3 y_2^4 + y_4 y_2 Y$, $x_2 = -a_i \tau_i \left|\mathbf{h}_i\right|^2$. Herein, $x_1 = a_i b_i$, $y_1 = -x_2 e^{2x_1} = a_i \tau_i \left|\mathbf{h}_i\right|^2 e^{2a_i b_i}$, $y_2 = e^{x_1}$, $y_3 = 1 + Y\left(-\frac{c_i}{e^{x_1}}\right)$, $y_4 = \frac{\left(\eta_{EE}^* \tau_i + \kappa\right)}{(T - \tau_i + \nu)} \frac{(T - \tau_i)\left(\sum_{k=i+1}^{I} P_k^u \left|\mathbf{h}_k\right|^2 + \sigma^2\right)}{\left|\mathbf{h}_i\right|^2} \frac{(\ln 2)}{B}$, $Y = \frac{M \left|\mathbf{h}_i\right|^2}{(T - \tau_i)\left(\sum_{k=i+1}^{I} P_k^u \left|\mathbf{h}_k\right|^2 + \sigma^2\right)}$.

Algorithm 1 Energy-Efficient Resource Allocation Based On Nonlinear Energy Harvesting

Input: Initialize the variables a_i, b_i, c_i, η_{EE}, $P_{\mathrm{bs,max}}$, $P_{\mathrm{uav,max}}$, $R_{\min}$, $\Delta\kappa$, $\Delta\lambda$, $\Delta\mu$, $\Delta\nu$, and ε.
while (not converge and $n \le N_{\max}$) **do**
  Update κ, λ, μ, ν according to (23) ~ (26).
  Calculate the p_i^t and τ_i by solving the formulas (22) and (21).
  if $R_{\mathrm{tot}}(p_i^t, \tau_i) - \eta_{\mathrm{EE}}^* E_{\mathrm{tot}}(p_i^t, \tau_i) \le \varepsilon$ **then**
    Convergence = true;
    return $p_i^{t*} = p_i^t, \tau_i^* = \tau_i$ and obtain η_{EE}^*;
  else
    Convergence = false;
    return $\eta_{\mathrm{EE}} = R_{\mathrm{tot}}(p_i^t, \tau_i)/E_{\mathrm{tot}}(p_i^t, \tau_i)$,
  end if
end while
Output: The optimal power p_i^{t*} and optimal time τ_i^*.

Table 1
Simulation parameters.

Parameter	Value	Parameter	Value	Parameter	Value
P_0	0.1	ρ	1.225	v_0	7.2
P_i	0.2	A	0.79	d_0	0.3
U_{tip}	200	s	0.05	$R_{\min}$	0.1
$P_{\mathrm{bs,max}}$	60 W	$P_{\mathrm{uav,max}}$	10 W	ε	10^{-7}
$N_{\max}$	1000	ϵ	0.2~0.9	a_i	150
I	5~10	c_i	0.024 W	b_i	0.014

The time for nonlinear EH can be obtained by solving Eq. (21) numerically. In this way, the optimal power and nonlinear EH time allocation for each iteration can be solved. Further, we use the downward gradient method to update the multipliers as follows.

$$\kappa(n+1) = [\kappa(n) - \Delta\,\kappa(P_{\mathrm{bs,max}} - p_i^t)]^+, \tag{23}$$

$$\lambda(n+1) = [\lambda(n) - \Delta\,\lambda(\tau_{\max} - \tau_i)]^+, \tag{24}$$

$$\mu(n+1) = [\mu(tn) - \Delta\,\mu(T - \tau_i)]^+, \tag{25}$$

$$\nu(n+1) = [\nu(n) - \Delta\,\nu(R_i - R_{\min})]^+, \tag{26}$$

where n is the iteration index, and $[\cdot]^+$ denotes the operation that takes the maximum between 0 and the variable. $\Delta\ \kappa$, $\Delta\ \lambda$, $\Delta\ \mu$ and $\Delta\ \nu$ are the step sizes. Based on the optimal value η_{EE}^* and the iteration update of the time allocation and power allocation parameters, the convergence can be obtained by satisfying the following $R_{\mathrm{tot}}\left(p_i^t, \tau_i\right) - \eta_{\mathrm{EE}}^* E_{\mathrm{tot}}\left(p_i^t, \tau_i\right) < \varepsilon$. Here, ε is a sufficiently small positive number. However, if the condition above cannot be achieved, $\eta_{\mathrm{EE}} = \frac{R_{\mathrm{tot}}(p_i^t, \tau_i)}{E_{\mathrm{tot}}(p_i^t, \tau_i)}$ will be updated until the convergence condition is satisfied. We summarize the power and nonlinear EH time allocation scheme in Algorithm 1.

4. Simulation results

In this section, we provide simulation results to verify the proposed resource allocation algorithm with the nonlinear EH model. We consider a multiantenna BS and several UAV users. The locations of the UAVs are generated according to uniformly random distribution within a range of (−50 m, 50 m) on the ground. In the simulations, the number of UAVs in the swarm is set to be no more than 10 and the number of antennas in the URA is set to be no more than 64 arranged as 2 × 2, 3 × 3, 6 × 6, 5 × 10 or 8 × 8 arrays. The antennas are placed in a rectangular region with 5 cm spacing between each antenna. Other simulation parameters [31] are listed in Table 1. For simplicity, we use "LEH" and "NEH" to denote linear EH and nonlinear EH, and denote the orthogonal multiple access scheme by "OMA" when describing and plotting the figures.

Fig. 2 shows the relationship between the transmit power and harvested power by the UAV for different EH models. We consider the linear EH model under four values of ϵ = 0.2, 0.3, 0.6, 0.8 and the nonlinear EH model. For the linear EH model [24], the harvested power by the UAV is linearly increased with the increase of the transmit power. The linear EH process is remarkably affected by the parameter ϵ, and the higher the parameter ϵ, the more power is harvested. For the nonlinear EH model, the harvested power by the UAV first increases and then flattens out. When the transmit power is over 50 W, the harvested power remains constant since the EH circuit enters into a saturation state.

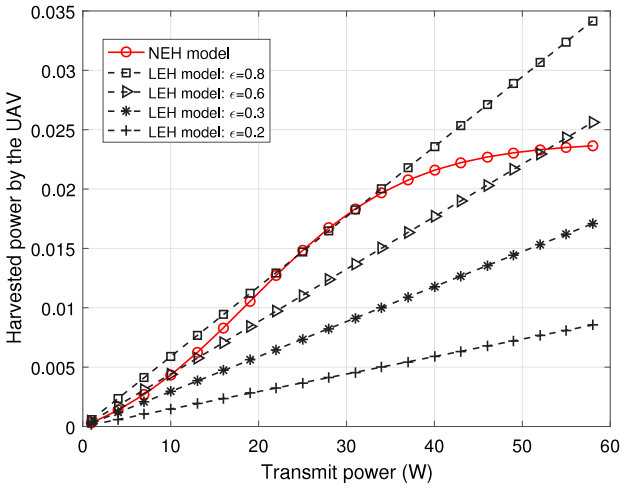


Fig. 2. Comparison of transmit power and harvested power conversion relationships of linear and nonlinear EH models.

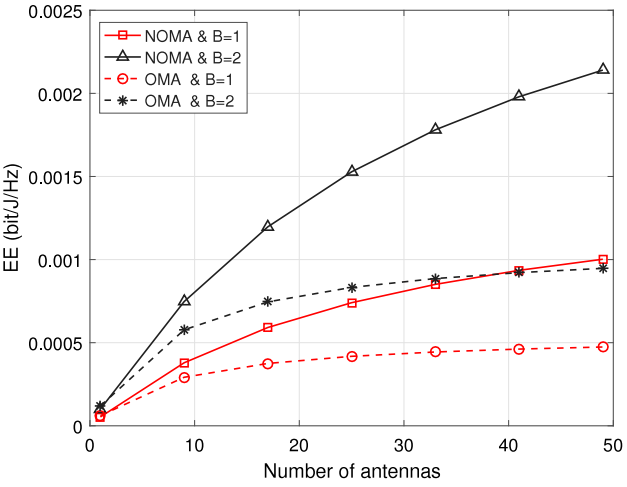


Fig. 3. The effect of antenna number on EE under different access schemes and bandwidths.

Fig. 3 shows the effect of antenna number on EE under NOMA and OMA access schemes. We can see that as the antenna number increases, EE increases and the trend of increase is gradually slowing down. This is because more antennas to some extent can help improve the channel capacity and thus EE. We also see that the proposed NOMA scheme outperforms the OMA scheme due to the adequacy of nonorthogonal resource allocation. In addition to the comparison of multiple access schemes, Fig. 3 also compares the EE of different bandwidth settings. We can see that under the same access scheme (orthogonal or nonorthogonal access mode), the greater the bandwidth, the higher the overall EE of the system. This shows that under the condition of sufficient bandwidth resources, appropriately increasing the channel bandwidth is helpful in improving the total EE of the system.

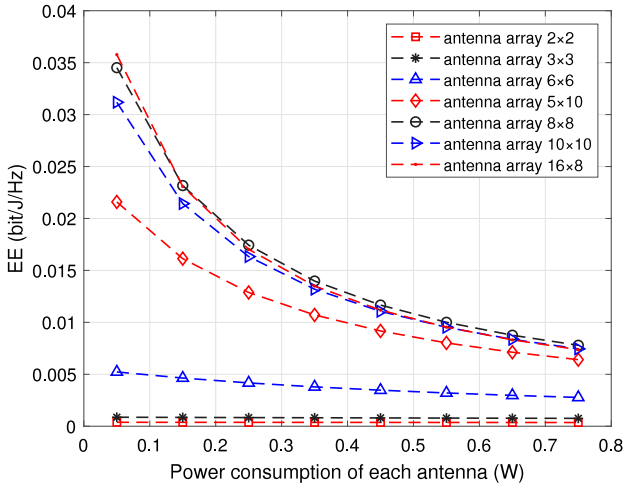


Fig. 4. EE under different antenna arrays.

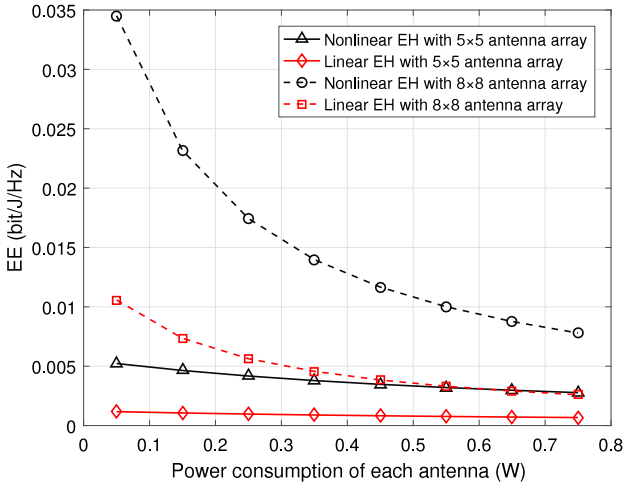


Fig. 6. Impact of linear and nonlinear EH models on EE with different antenna arrays.

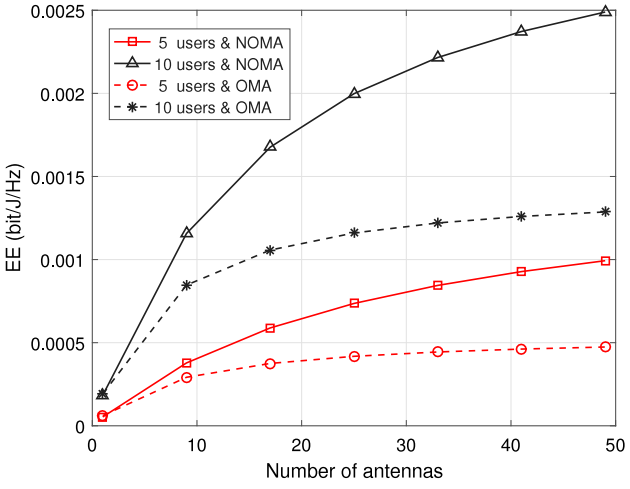


Fig. 5. EE under different user number and access schemes.

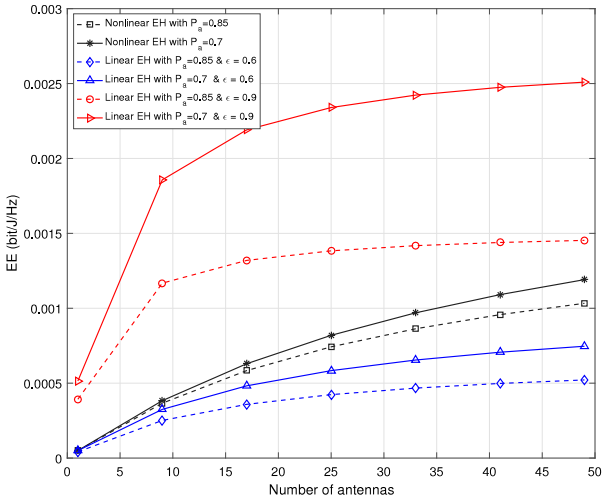
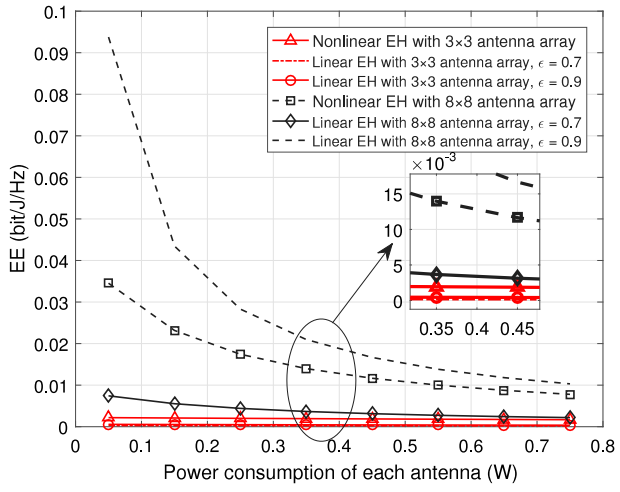


Fig. 7. Impact of linear and nonlinear EH models on EE with different antenna power consumption.

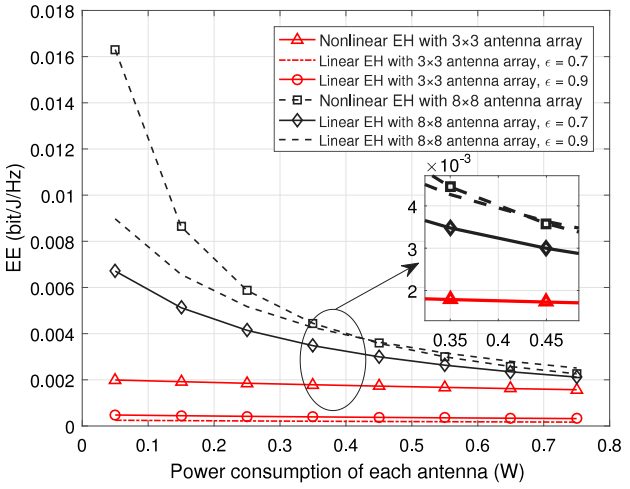
Fig. 4 investigates the impact of antenna arrays on the EE. We consider seven antenna array cases: 2 × 2 array, 3 × 3 array, 6 × 6 array, 5 × 10 array, 8 × 8 array, 10 × 10 array, and 16 × 8 array. We can observe that as the power consumption of each antenna increases, the EE for all the cases is gradually decreased. This is because too much power consumption of each antenna will increase the total energy consumption, thereby reducing the EE. In addition, we can see that the system EE of 8 × 8 antenna array is superior to that of the other six antenna arrays. This shows that the number of antenna arrays is not the more the better, and this phenomenon is caused by the limited energy accumulation induced by nonlinear energy harvesting and the huge power loss overhead of massive MIMO antennas and other practical factors. This means more antennas promote greater EE when the antenna number is controlled within a reasonable range.

Fig. 5 shows the EE versus antenna number under different user numbers and access schemes. As the antenna number increases, the EE first fast increases and then slightly climbs to a maximum and keeps constant, which is consistent with Fig. 3. We also compare the impact of user numbers and access schemes on the EE, and choose 5 users, 10 users and orthogonal and non-orthogonal multiple access modes respectively. According to the Fig. 5, the proposed algorithm with 10 users under the NOMA scheme is superior to the other cases. This shows that the NOMA scheme can allow more users to access wireless communication networks under the condition of sufficient power distribution, and gain greater spectrum efficiency and EE.

Fig. 6 presents the impact of linear and nonlinear EH models on the EE. We observe that as the power consumption of each antenna increases from 0.1 W to 0.8 W, the EE for all the cases decreases, which is consistent with Fig. 4. Moreover, we compare the curves of linear and nonlinear energy harvesting models with 5 × 5 array and 8 × 8 array. We note that the EE under nonlinear EH model is superior to that under linear EH model. This is because the linear EH model uses a fixed power conversion efficiency, which brings about a mismatch in resource allocation. The nonlinear EH model is more practical due to its saturation characteristic of the receiver EH circuit. In addition, compared with the other curves, the nonlinear model with 8 × 8 antenna array presents the best EE values. This is because with the help of large-scale antennas, the spectral efficiency of the system has been significantly improved, and thus the energy efficiency of the system has been increased.



(a) Case I: 5 Users



(b) Case II:10 Users

Fig. 8. Impact of linear and nonlinear EH models on EE with different antenna arrays and energy conversion efficiency.

To demonstrate the impact of energy harvesting models on EE based on our proposed algorithm, the EE with respect to the number of antenna under various antenna power consumption is plotted in Fig. 7. Therein, the EE of the linear EH model is presented by different antenna powers (i.e., $P_a = 0.85$ and $P_a = 0.7$), which is closely related to the energy conversion efficiency ϵ. We observe that the EE of the linear EH model with $\epsilon = 0.9$ is significantly higher than the other scheme with $\epsilon = 0.6$, as less energy is lost in the collection process. On the contrary, with more antenna power loss, the EE is lower than other cases. For example, in both linear and nonlinear EH models, the EE corresponding to antenna power $P_a = 0.7$ is better than that corresponding to antenna power $P_a = 0.85$. In addition, we can observe that when the conversion efficiency of the linear EH model is high, its EE performance is better than that of the nonlinear EH model, which indicates that the extremely high conversion efficiency can alleviate the weakening degree of the system EE of the non-actual EH model.

Fig. 8 illustrates the system EE in the proposed WPT-powered UAV NOMA scheme when both the EH model and the number of antennas vary under two cases of 5 users and 10 users. As seen in Fig. 8, the greater power consumption per UAV antenna can result in lower system EE for any situations. The reason is that, in our proposed scheme, the greater the power consumption per antenna, the greater the power consumption of the system, so the smaller the amount of information per unit power consumption, that is, the smaller the EE. By observing the change of system EE under the number of 5 users in Fig. 8(a), we can see that the EE of linear EH model with 64 antennas and $\epsilon = 0.9$ shows the best performance than other five curves. It is worth noting that in the case of serving more than 10 users in Fig. 8(b), we can observe that nonlinear EH model with 64 antennas can attain the best EE performance relative to other five curves. This is because with the increase of service users, the gap between the linear EH model and the actual circuit EH model increases, and the increasing number of users leads to an excessive circuit loss burden, leading to the gradual convergence and approximation of the system EE values of the linear and nonlinear models.

5. Conclusion

In this paper, we proposed a NOMA-based WPT-powered UAV communication system under a practical nonlinear energy harvesting model. To cater to the massive connectivity and improve the system EE, an energy-efficient resource allocation problem was formulated. As the problem is nontrivial and nonconvex, we converted the original fractional problem to a subtractive form. Then we employed the Lagrange dual decomposition approach to solve the transformed problem via several iterations. Simulation results showed that the EE of the proposed energy-efficient resource allocation under the nonlinear EH model is more consistent with the actual circuit state than that under the linear EH model alternative.

In UAV-assisted communication systems, the uses of WPT and nonlinear EH-based techniques have become quite popular to provide enhanced coverage to ground users and required power to energy-constrained low power devices [32]. There are many promising directions worth researching in the future. For instance, the integration of MEC into nonlinear EH-based UAV communication networks is an important and promising research direction that can effectively improve the overall performance of systems with mission-critical and computation-intensive tasks [33]. Intelligent reflecting surface (IRS)-assisted UAV communication with nonlinear EH model is also an important research issue, which can overcome the limitations, control the wireless propagation environment, and improve the system performance of energy-constrained UAV communications [34]. Moreover, it is worth extending our work to more energy-efficient scenarios, such as cell-free massive multiple input multiple output (MaMIMO)-based nonlinear EH-enabled UAV framework [35], and integrated sensing UAV communication networks with nonlinear EH [36], etc.

CRediT authorship contribution statement

Zhongyu Wang: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing. **Yashuai Cao:** Writing – original draft, Writing – review & editing. **Zheng Chang:** Investigation. **Tiejun Lv:** Investigation. **Wei Ni:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Appendix. Derivation of Theorem 1

A.1. Sufficiency

To facilitate the derivation, we define $\mathcal{P}$ and $\mathcal{T}$ as the set of feasible solutions for **P1** in (13), and $\eta_{\mathrm{EE}}\left(p_i^t,\tau_i\right)=R_{\mathrm{tot}}\left(p_i^t,\tau_i\right)/E_{\mathrm{tot}}\left(p_i^t,\tau_i\right)$. Without loss of generality, η_{EE}^*, $p_i^{t*}\in\mathcal{M}$, $\tau_i^*\in\mathcal{Q}$, are defined as the optimal EE, power allocation, WPT time division policies of the primal problem **P1**, respectively. We can obtain the optimal EE as

$$\eta_{\mathrm{EE}}^*=\frac{R_{\mathrm{tot}}\left(p_i^{t*},\tau_i^*\right)}{E_{\mathrm{tot}}\left(p_i^{t*},\tau_i^*\right)}\geq\frac{R_{\mathrm{tot}}\left(p_i^t,\tau_i\right)}{E_{\mathrm{tot}}\left(p_i^t,\tau_i\right)},\quad\forall\ p_i^t\in\mathcal{P},\tau_i\in\mathcal{T}. \tag{A.1}$$

Since the denominator of (A.1) is positive, according to the transformation of the equation and inequality, we have

$$\begin{cases}R_{\mathrm{tot}}\left(p_i^t,\tau_i\right)-\eta_{EE}^*E_{\mathrm{tot}}\left(p_i^t,\tau_i\right)\leq 0,\\ R_{\mathrm{tot}}\left(p_i^{t*},\tau_i^*\right)-\eta_{EE}^*E_{\mathrm{tot}}\left(p_i^{t*},\tau_i^*\right)=0.\end{cases} \tag{A.2}$$

We conclude that $\max_{p,\tau}R_{\mathrm{tot}}\left(p_i^t,\tau_i\right)-\eta_{\mathrm{EE}}E_{\mathrm{tot}}\left(p_i^t,\tau_i\right)=0$ is achievable and feasible at the optimal transmit power, WPT time allocation p_i^{t*},τ_i^*.

A.2. Necessity

The converse implication of Theorem 1 can be proved as follows. $p_i^{t\prime}$ and τ_i' are assumed as the optimal power and WPT time policies of the equivalent objective function so as to satisfy the following condition

$$R_{\mathrm{tot}}\left(p_i^{t\prime},\tau_i'\right)-\eta_{EE}^*E_{\mathrm{tot}}\left(p_i^{t\prime},\tau_i'\right)=0\quad. \tag{A.3}$$

Given any feasible power and time policies, $p_i^t\in\mathcal{P}$ and $\tau_i\in\mathcal{T}$, we have

$$\begin{aligned}&R_{\mathrm{tot}}\left(p_i^t,\tau_i\right)-\eta_{EE}^*E_{\mathrm{tot}}\left(p_i^t,\tau_i\right)\\ &\leq R_{\mathrm{tot}}\left(p_i^{t\prime},\tau_i'\right)-\eta_{EE}^*E_{\mathrm{tot}}\left(p_i^{t\prime},\tau_i'\right)=0.\end{aligned} \tag{A.4}$$

The inequality (A.4) yields

$$\begin{aligned}&\frac{R_{\mathrm{tot}}\left(p_i^t,\tau_i\right)}{E_{\mathrm{tot}}\left(p_i^t,\tau_i\right)}\leq\eta_{\mathrm{EE}}^*,\forall\ p_i^t\in\mathcal{P},\tau_i\in\mathcal{T},\\ &\frac{R_{\mathrm{tot}}\left(p_i^{t\prime},\tau_i'\right)}{E_{\mathrm{tot}}\left(p_i^{t\prime},\tau_i'\right)}=\eta_{\mathrm{EE}}^*.\end{aligned} \tag{A.5}$$

This indicates that the optimal solutions $p_i^{t\prime}$ and τ_i' for the objective function corresponds to the optimal resource allocation policies for the original objective **P1**.

The proof of Theorem 1 is completed. ■

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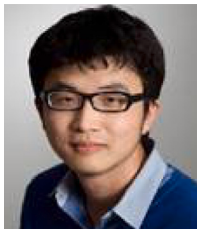
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